

Fast Monte Carlo Pricing of Asian Options under Rough Fractional Stochastic Volatility

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Abstract

We study Monte Carlo pricing of a fixed-strike arithmetic Asian call option under the Rough Fractional Stochastic Volatility (RFSV) model [9], in which log-volatility follows a fractional Brownian motion with Hurst exponent $H = 0.10$. The non-Markovian structure of fBM forces every sample path to be drawn from a dense $N \times N$ Gaussian distribution, requiring an $O(N^3)$ Cholesky setup and an $O(MN^2)$ per-path simulation cost. We implement and benchmark an exact dense Cholesky baseline against two algorithms that exploit structure in the covariance matrix: a circulant-embedding FFT method ($O(N \log N)$ per path, exploiting the stationarity of fBM increments), and a global randomized SVD method ($O(Nk)$ per path, exploiting low-rank structure). We show that the circulant embedding is exact for every $H \leq 1/2$, with smallest eigenvalue $\Delta t^{2H} (N^{2H} - (N-1)^{2H})$ in closed form. Over $N \in \{252, 500, 1000\}$ the Cholesky and FFT costs scale as $N^{1.33}$ and $N^{0.95}$, but at $N = 1000$ the FFT method is only $\approx 9\%$ faster than a triangular Cholesky baseline, because random number generation rather than the covariance factor dominates its per-path cost. The rank-32 rSVD method is $2.7\times$ faster than Cholesky at $N = 1000$, but its covariance error decays slowly, from 1.9% at $k = 32$ to 1.2% at $k = 128$, because the rough kernel's singularity lies on the diagonal, where a global low-rank approximation cannot isolate it.

1 Introduction

1.1 The option contract

I am interested in pricing fixed-strike, arithmetic-average Asian call options. The payoff at expiry T is

$$C(T) = \max(A(0, T) - K, 0), \quad A(0, T) = \frac{1}{N} \sum_{j=1}^N S_{t_j}, \quad (1)$$

where S_{t_j} is the price of the underlying at time $t_j = jT/N$ and K is the strike. Path-dependence through the average A prevents the Black–Scholes machinery from yielding an exact closed form, making approximations or simulation necessary. Lévy [15] proposed a moment-matching approximation: matching the first two moments of the discrete arithmetic average to a lognormal yields a tractable formula that is accurate under constant-volatility Black–Scholes dynamics, where S_t is lognormal.

The situation is more difficult under Rough Fractional Stochastic Volatility (RFSV) [9], in which $\log \sigma_t = \nu W_t^H$, where W_t^H is a fractional Brownian motion with Hurst exponent $H \approx 0.10$. Under RFSV, S_t is no longer lognormal, and the two-moment Lévy approximation cannot capture the heavy skew and fat tails introduced by a large vol-of-vol ν . Computing even the second joint moment $\mathbb{E}[S_{t_i} S_{t_j}]$ analytically is intractable: it requires the expectation of the integrated volatility path, which depends on the full continuous history of W^H up to each time point.

Monte Carlo sidesteps the distributional intractability by estimating the expectation directly from simulated paths. By the law of large numbers,

$$\frac{1}{M} \sum_{m=1}^M e^{-rT} \max(A^{(m)} - K, 0) \xrightarrow{M \rightarrow \infty} \mathbb{E}^{\mathbb{Q}}[e^{-rT} \max(A - K, 0)], \quad (2)$$

where $\mathbb{Q}$ is the risk-neutral measure and $A^{(m)}$ is the arithmetic average along the m -th simulated path. No closed-form density or moment-matching is required; we need only the ability to simulate correlated log-volatility paths.

1.2 Why rough volatility

Gatheral et al. [9] provide compelling empirical evidence that equity log-realized-variance behaves as fractional Brownian motion with $H \approx 0.10$. Estimating H by ordinary least squares on log-realized-variance increments across five major indices (S&P 500, NASDAQ, DAX, FTSE 100, Nikkei 225) and time scales ranging from one day to one month, they find a consistent Hurst exponent well below $1/2$, with $R^2 > 0.90$ in each case. I take $H = 0.10$ and $\nu = 0.30$ from their estimates on high-frequency realized variance data [18]. One caveat concerns units: Gatheral et al. [9] measure time in days, while my engine measures it in years ($T = 1$). In year units the same fit corresponds to $\nu \approx 0.3 \cdot 252^{0.1} \approx 0.52$,

so my experiments run at a lower vol-of-vol than the empirical estimate. My own variogram fit to non-overlapping 5-day realized variance from `yfinance`, with time in years, gives $H \approx 0.12$ ($R^2 \approx 0.98$ across a handful of lags) and $\nu \approx 0.87$, the latter inflated by measurement noise in weekly squared-return variance.

Standard Brownian motion corresponds to $H = 1/2$, with uncorrelated increments. At $H = 0.10$, volatility paths are far rougher than BM: increments are negatively correlated at all lags, the path is less regular than classical stochastic volatility models, and the realized variance process mean-reverts on short timescales. The short-dated implied volatility smile steepens beyond what classical stochastic volatility models expect, and volatility clustering manifests on timescales an order of magnitude shorter than, e.g., Heston.

The RFSV model I studied is [9]

$$\log \sigma_t = \nu W_t^H, \quad dS_t = rS_t dt + \sigma_t S_t dW_t^{\mathbb{Q}}, \quad (3)$$

where W_t^H is the fractional Brownian motion of Mandelbrot and Van Ness [16] with covariance $\mathbb{E}[W_t^H W_s^H] = \frac{1}{2}(t^{2H} + s^{2H} - |t - s|^{2H})$. I set $r = 0$ throughout for simplicity (consistent with `params.hpp`); the drift term drops out of all pricing formulas but is retained in the SDE for completeness. I use $H = 0.10$ and $\nu = 0.30$ throughout, and also explore the sensitivity of option prices to both parameters in Section 6.3.

1.3 The computational bottleneck

Naïve Monte Carlo for RFSV requires $O(N^3)$ setup work. Because fBM is non-Markovian, there is no forward recursion: the value of W_t^H depends on the entire history of the driving noise, not just its most recent value. Generating each sample path therefore requires drawing from the full N -dimensional Gaussian $\mathcal{N}(0, C)$, where C is a dense $N \times N$ positive-definite matrix with $C_{ij} = \frac{1}{2}(t_i^{2H} + t_j^{2H} - |t_i - t_j|^{2H})$. The standard approach factorizes $C = LL^\top$ via Cholesky decomposition once ($O(N^3/3)$ flops), then generates each path by computing νLz for $z \sim \mathcal{N}(0, I)$ ($O(N^2)$ flops per path, a lower-triangular matrix-vector multiply).

At $N = 252$ trading days and $M = 10,000$ paths the costs are concrete. Cholesky decomposition requires exactly $N^3/3 \approx 5.3$ million floating-point operations (the leading constant $1/3$ compares favourably to $2/3$ for LU); this is a one-time cost. The per-path triangular multiply costs $O(N^2) \approx 63,000$ flops, so the total Monte Carlo loop costs $O(MN^2) \approx 630$ million flops, more than $100\times$ the factorization. MC pricing error falls as $O(M^{-1/2})$: with $\sigma_{\text{payoff}} \approx 61$ (Section 5.3), $M = 10,000$ paths give a standard error of ≈ 0.6 price units, about 2.6% of the at-the-money price, and halving it requires four times as many paths.

Trading desks building sensitivity surfaces run parameter sweeps over H , ν , and K , easily multiplying M by factors of 10–100. Intraday risk monitoring at $N \approx 19,656$ (five-minute intervals over 252 trading days of 6.5 hours each: $252 \times 78 = 19,656$) increases the per-path cost by $\sim 6,000\times$ relative to daily steps.

1.4 Approach and scope

I implement and benchmark three algorithms that reduce the cost of fBM path simulation by exploiting structure in the covariance matrix C :

1. **Dense Cholesky** ($O(N^2)/\text{path}$, exact): exploits the positive-definiteness of C to factor it exactly. By “exact simulation” I mean that the method is free of time-discretization bias: it samples the correct fBM distribution for any finite N , unlike Euler–Maruyama approximations. Serves as the correctness baseline against which the other methods are validated.
2. **Circulant embedding & FFT** ($O(N \log N)/\text{path}$): exploits the fact that the increments of fBM (fractional Gaussian noise, fGn) are stationary. Their covariance matrix is Toeplitz, which can be embedded into a circulant matrix, which can be diagonalized by the DFT [6, 19]. For $H \leq 1/2$ the embedding is guaranteed positive [5], so the method is exact on the grid.
3. **Global low-rank rSVD** ($O(Nk)/\text{path}$): exploits the low-rank structure of smooth off-diagonal blocks of C : the off-diagonal blocks admit low-rank approximations. A randomized SVD [13] compresses $C \approx U \text{diag}(S) U^\top$ at rank k , reducing per-path cost from $O(N^2)$ to $O(Nk)$. In practice the implementation is a global approximation of the entire $N \times N$ matrix (including the singular near-diagonal region) rather than of the smooth off-diagonal blocks alone. This is blunter, and must allocate rank budget to the diagonal singularity, which is why the covariance error is still 1.2% at $k = 128$ for $H = 0.10$ (§5.5).

Section 2 develops the mathematical properties of C that each method exploits. Section 3 describes each algorithm and its complexity. Sections 5 and 6 present empirical timing, memory, accuracy, and model validation results.

The project does not address variance reduction (quasi-Monte Carlo, control variates), model extensions beyond fBM-RFSV (rough Bergomi, multifactor), or GPU parallelism, though Section 7 situates each of these as a natural next step.

2 Mathematical Foundations

2.1 Fractional Brownian motion and its covariance structure

Fractional Brownian motion W_t^H is the unique (up to scaling) continuous Gaussian process with stationary increments and the self-similarity property $W_{ct}^H \stackrel{d}{=} c^H W_t^H$ for all $c > 0$ [16]. Its covariance kernel is

$$\mathbb{E}[W_t^H W_s^H] = \frac{1}{2} (t^{2H} + s^{2H} - |t - s|^{2H}), \quad (4)$$

which is symmetric, positive-definite for $H \in (0, 1)$, and reduces to the standard Brownian covariance $\min(t, s)$ at $H = 1/2$. The self-similarity $W_{ct}^H \stackrel{d}{=} c^H W_t^H$ and

the stationarity of increments $W_{t+h}^H - W_t^H \stackrel{d}{=} W_h^H$ are the properties that make fBM a tractable model of rough volatility; §2.2 exploits increment stationarity directly. The critical property for simulation is that fBM is non-Markovian for $H \neq 1/2$: the value of W_t^H depends on the entire history of the driving noise, not just its most recent value. This rules out any forward recursion and forces each sample path to be drawn from a joint N -dimensional Gaussian.

Discretizing $[0, T]$ into N equally spaced points $t_j = jT/N$ yields an $N \times N$ covariance matrix C with $C_{jk} = \mathbb{E}[W_{t_j}^H W_{t_k}^H]$ as in equation (4). The matrix C is dense (every pair of time points is correlated) and positive-definite for all $H \in (0, 1)$ and $N \geq 1$. This density explains the $O(N^2)$ per-path cost: sampling from $\mathcal{N}(0, \nu^2 C)$ by any exact method requires touching all $N^2/2$ entries of C . Algorithm 1 (§3.1) accepts this cost and factorizes C exactly; Algorithms 2 and 3 avoid it by exploiting structure that C possesses.

2.2 Stationarity of fGn and the Toeplitz property

Although the fBM covariance matrix is dense, its increment process has a crucial structural simplification. Define the fractional Gaussian noise (fGn) increments $\delta W_m = W_{(m+1)\Delta t}^H - W_{m\Delta t}^H$ for $m = 0, \dots, N-1$, where $\Delta t = T/N$. Stationarity of fBM increments implies

$$\gamma(k) := \text{Cov}(\delta W_m, \delta W_{m+k}) = \frac{\Delta t^{2H}}{2} (|k+1|^{2H} + |k-1|^{2H} - 2|k|^{2H}), \quad (5)$$

which depends only on the lag $k = |m - n|$, not on the absolute index m [19]. The $N \times N$ covariance matrix of $(\delta W_0, \dots, \delta W_{N-1})$ is therefore Toeplitz: its (m, n) entry equals $\gamma(|m - n|)$, so every diagonal is constant.

Figure 1 illustrates the contrast. Panels (a) and (b) show the $N \times N$ covariance heatmaps of fBM and fGn respectively. The fBM matrix is not Toeplitz: row m of C varies with m because the variance t_m^{2H} grows with time. The fGn matrix has constant diagonals, confirming the Toeplitz property exactly. Panel (c) shows diagonal slices $C[m, m+k]$ vs. lag k for two different rows m : the fGn curves overlap to numerical precision, while the fBM curves diverge.

Algorithm 2 (§3.2) depends on this Toeplitz structure: only Toeplitz matrices embed into a circulant diagonalizable by the discrete Fourier transform. Applying the FFT method directly to the fBM covariance matrix produces negative eigenvalues and pathological path output.

2.3 Circulant embedding and its positivity condition

A Toeplitz matrix T with first row $(\gamma(0), \gamma(1), \dots, \gamma(N-1))$ can be embedded into a $2N \times 2N$ circulant matrix C_{circ} whose first row is the symmetric extension

$$c = (\gamma(0), \gamma(1), \dots, \gamma(N-1), 0, \gamma(N-1), \dots, \gamma(1)). \quad (6)$$

Circulant matrices are diagonalized by the DFT: the eigenvalues of C_{circ} are exactly $\lambda = \text{FFT}(c)$, computable in $O(N \log N)$ operations. To sample, draw

$W = a + ib$ with $a, b \sim \mathcal{N}(0, I_{2N})$ independent (so each component has real and imaginary parts that are independent standard normals), scale component-wise by $\sqrt{\lambda_j/(2N)}$, apply the inverse FFT, and take the real part of the first N elements [6]. One forward FFT (once per experiment) to compute $\{\lambda_j\}$, then one inverse FFT per path: total $O(N \log N)$ per path.

The method is valid only if every eigenvalue λ_j is non-negative, so that $\sqrt{\lambda_j}$ is real [19]. For rough H this holds at every N , not just asymptotically. Craigmile [5] proves that the Davies–Harte embedding is non-negative definite for any stationary process whose autocovariance is non-positive at all non-zero lags, a class that includes fGn with $H < 1/2$. The argument is short enough to give in full for the embedding used here. Because $x \mapsto x^{2H}$ is concave for $H < 1/2$, the second difference in equation (5) is negative, so $\gamma(k) < 0$ for every $k \geq 1$. The eigenvalues of the symmetric circulant are

$$\lambda_j = \gamma(0) + 2 \sum_{k=1}^{N-1} \gamma(k) \cos\left(\frac{\pi j k}{N}\right), \quad j = 0, \dots, 2N-1, \quad (7)$$

and because each $\gamma(k)$ is negative, replacing every cosine by 1 can only lower the sum. The smallest eigenvalue is therefore λ_0 , and the sum telescopes:

$$\begin{aligned} \lambda_{\min} = \lambda_0 &= \gamma(0) + 2 \sum_{k=1}^{N-1} \gamma(k) \\ &= \Delta t^{2H} (N^{2H} - (N-1)^{2H}) \approx 2H \Delta t^{2H} N^{2H-1} > 0. \end{aligned} \quad (8)$$

The FFT method is thus exact on the grid for every $H \leq 1/2$ (at $H = 1/2$ the increments are white noise and every $\lambda_j = \gamma(0)$). The C++ pricer checks the sign of every eigenvalue and throws rather than clipping.

Equation (8) also shows that the margin is thin. Relative to the increment variance $\gamma(0) = \Delta t^{2H}$, the smallest eigenvalue is $N^{2H} - (N-1)^{2H}$: 0.24% at $H = 0.10$, $N = 252$ and 0.08% at $N = 1000$, decaying like $N^{-(1-2H)}$. This mirrors the fGn spectral density, which vanishes at zero frequency when $H < 1/2$. Section 5.4 confirms the formula numerically. The key pitfall is $\gamma(0)$ itself: dropping the $|k-1|^{2H}$ term at $k=0$ halves it, which breaks the bound and spuriously makes 28% of the eigenvalues negative at $H = 0.10$. An earlier version of my stability analysis made exactly this mistake and concluded, wrongly, that the FFT method needed eigenvalue clipping.

2.4 Low-rank structure of smooth off-diagonal blocks

The fBM kernel $C(t, s) = \frac{1}{2}(t^{2H} + s^{2H} - |t-s|^{2H})$ has a singularity at $t = s$, since the term $|t-s|^{2H}$ introduces one. For $H = 0.10$ this gives $|t-s|^{0.2}$, whose derivative $0.2|t-s|^{-0.8}$ is unbounded as $t \rightarrow s$. The function is Hölder-continuous with exponent 0.2 but strictly non-differentiable on the diagonal.

Off-diagonal blocks of C (submatrices coupling two disjoint time intervals) lie in the smooth region $t \neq s$. Smooth bivariate kernels generate matrices with rapidly decaying singular values on such blocks [4].

Table 1: Complexity of the three fBM path generation methods. N : time steps; M : paths; k : rSVD target rank. “Exact” means the method samples the correct Gaussian distribution on the grid, up to floating-point error; for the FFT method this rests on the positivity result of §2.3.

Method	Setup	Per path	Memory	Accuracy
Dense Cholesky	$O(N^3)$	$O(N^2)$	$O(N^2)$	Exact
Circulant + FFT	$O(N \log N)$	$O(N \log N)$	$O(N)$	Exact
Low-rank rSVD	$O(N^2 k)$	$O(Nk)$	$O(Nk)$	Approximate

Figure 1(d) confirms this, and locates the difficulty precisely. The off-diagonal block coupling the first half of the grid to the second half is exactly rank 1 at $H = 0.50$, where the kernel is $\min(t, s)$. At $H = 0.10$ its singular values still fall geometrically: below 1% of σ_1 by rank $k = 3$ and below 10^{-4} by $k = 5$. The spectrum of the full matrix C behaves very differently. It decays only algebraically, and more slowly at $H = 0.10$ than at $H = 0.50$. The slow decay that limits a global low-rank approximation therefore comes from the near-diagonal singularity, not from the far field. This is exactly the structure a hierarchical matrix exploits (§7.1).

3 The three algorithms

Table 1 summarizes the three methods. The rest of this section describes each in turn.

3.1 Algorithm 1: Dense Cholesky

The fBM covariance matrix C is symmetric positive-definite for all $H \in (0, 1)$ and $N \geq 1$, which is sufficient for an exact Cholesky factorization $C = LL^\top$.

Algorithm

1. Build $C \in \mathbb{R}^{N \times N}$ analytically from equation (4): $O(N^2)$ flops.
2. Factorize $C = LL^\top$ in place via Cholesky decomposition [11]: $N^3/3$ flops. This is a one-time cost amortized over all M paths.
3. For each path: draw $z \sim \mathcal{N}(0, I_N)$ and compute νLz , a lower-triangular matrix-vector multiply (N^2 flops), which gives the fBM path directly; then simulate the stock price path and accumulate the payoff.

The method is exact. The cost is $O(N^2)$ memory for L (7.6 MB at $N = 1000$, which still fits in the M2’s 16 MB L2 cache) and $O(MN^2)$ total work, which dominates at practical M . This method serves as the correctness baseline: prices from the other two methods are validated against it. For the comparison to

be fair, the baseline must do only the work it needs. An earlier version of my implementation multiplied by a dense copy of L , zeros included, at $2N^2$ flops per path; §4.6 describes the fix and its effect.

3.2 Algorithm 2: Circulant embedding + FFT (Davies-Harte)

The fBM covariance is not Toeplitz, but the covariance of the fGn increments δW_k is (§2.2). A Toeplitz matrix embeds into a circulant; circulant matrices are diagonalized by the DFT (§2.3).

Algorithm

1. Compute the fGn autocovariance $\gamma(0), \dots, \gamma(N-1)$ and form the symmetric circulant first row $c \in \mathbb{R}^{2N}$.
2. Forward FFT: $\lambda = \text{FFT}(c)$; the eigenvalues are all real (the circulant is symmetric) and positive (§2.3). The code checks the sign and throws if any is negative. $O(N \log N)$ flops, computed once.
3. Pre-compute $s_j = \sqrt{\lambda_j/(2N)}$ for all j .
4. For each path: draw $W \in \mathbb{C}^{2N}$ whose real and imaginary parts are i.i.d. $\mathcal{N}(0, 1)$, scale $W_j \leftarrow s_j W_j$, apply inverse FFT, take the real part of the first N entries to obtain fGn increments, cumsum to fBM. $O(N \log N)$ flops per path.

FFTW plans for the forward and inverse transforms are created once before the Monte Carlo loop and reused across all M paths [8].

No clipping is needed. By the argument of §2.3, every eigenvalue is positive for $H \leq 1/2$, so the method is exact on the grid at any N [5]. The smallest eigenvalue is 0.24% of $\gamma(0)$ at $H = 0.10$, $N = 252$ (§5.4): small, but positive by construction.

3.3 Algorithm 3: Global low-rank rSVD

Away from the diagonal $t = s$, the fBM kernel is smooth, so off-diagonal blocks of C are well-approximated at low rank (§2.4). A global rank- k approximation $C \approx U \text{diag}(S) U^\top$ replaces C with a compact factor $L_k = U \text{diag}(\sqrt{S}) \in \mathbb{R}^{N \times k}$, dropping the per-path cost from $O(N^2)$ to $O(Nk)$.

Algorithm

1. Build $C \in \mathbb{R}^{N \times N}$ analytically: $O(N^2)$ flops.
2. Compute a rank- $(k + p)$ randomized SVD of C with $q = 2$ rounds of subspace iteration, re-orthonormalizing after every product (Algorithm 4.4 of Halko et al. [13]; see §4.5): $O(N^2 k)$ flops. Retain the top- k singular triplets $(U, S, U^\top)$.

3. Form the approximate Cholesky factor $L_k = U \text{diag}(\sqrt{S}) \in \mathbb{R}^{N \times k}$: $L_k L_k^\top \approx C$.
4. For each path: draw $z \sim \mathcal{N}(0, I_k)$, compute $\nu L_k z$ ($O(Nk)$ flops), cumsum to approximate fBM, simulate payoff.

What it sacrifices. Exactness. The sampled paths have covariance $\nu^2 C_k$ rather than $\nu^2 C$, with the error controlled by rank k . Section 5.5 shows that the relative Frobenius error falls slowly, from 8.5% at $k = 2$ to 1.2% at $k = 128$, because the spectrum of the full matrix decays only algebraically (Figure 1(d)).

This is a global rSVD, a single rank- k approximation of the entire $N \times N$ matrix. It must allocate rank budget to both the smooth far-off-diagonal blocks and the singular near-diagonal region, which drives up the required k at small H . A true hierarchical H-matrix keeps near-diagonal blocks dense and compresses only far-off-diagonal blocks [12].

3.4 Other algorithms for further exploration

True hierarchical H-matrix A recursive block-tree H-matrix decomposes C into $O(\log N)$ levels of blocks: near-diagonal blocks at each level are kept dense, while well-separated blocks at the same level are approximated at rank proportional to their distance from the diagonal. The resulting structure supports an H-Cholesky factorization with $O(Nk \log N)$ construction cost and $O(Nk)$ per-path cost, which is better scaling than the global rSVD at large N [12, 2]. The complexity gain is a $\log N$ factor over Algorithm 3; it requires implementing a recursive block-tree and the associated H-matrix arithmetic (§7.1).

Hybrid Scheme for rough Bergomi The Hybrid Scheme of Bennedsen et al. [3] achieves high RMSE reduction over naïve Euler discretization at the same $O(N \log N)$ cost. It does so by exploiting the separability of the Volterra kernel $K(t, s) = (t - s)^{H-1/2}$ near $t = s$ into a singular analytic part and a smooth remainder. However, this kernel governs the rough Bergomi model [1], not the fBM-RFSV model studied here: log-volatility in rough Bergomi is $\int_0^t (t - s)^{H-1/2} dW_s$, a Volterra integral, whereas in RFSV it is simply νW_t^H . Adopting the Hybrid Scheme would require switching model class.

Quasi-Monte Carlo Replacing pseudo-random Gaussian draws with low-discrepancy sequences [17] reduces MC standard error from $O(M^{-1/2})$ to near $O(M^{-1})$, an improvement that can be combined with any of the three algorithms.

4 Implementation Notes

This section documents the non-obvious design decisions in the C++ codebase that are not fully captured by the complexity table in §3.

4.1 fGn autocovariance, not fBM covariance

The circulant embedding requires a stationary covariance sequence. fBM itself is non-stationary: the variance $\mathbb{E}[(W_t^H)^2] = t^{2H}$ grows with t , so row i of the fBM covariance matrix C is different from row $j \neq i$. Attempting to embed C into a circulant and taking the FFT produces a mix of positive and negative eigenvalues even for $H = 1/2$; the method collapses entirely.

The correct input is the fGn autocovariance $\gamma(k)$ of increments (§2.2):

$$\gamma(k) = \frac{\Delta t^{2H}}{2} ((k+1)^{2H} + (k-1)^{2H} - 2k^{2H}), \quad (9)$$

which depends only on lag k . The pricer uses the fGn autocovariance to build the symmetric circulant first row $c = (\gamma(0), \dots, \gamma(N-1), 0, \gamma(N-1), \dots, \gamma(1))$ and then cumsums the resulting fGn increments back into fBM log-volatility. The bug is silent: the program runs and produces plausible-looking paths, just with the wrong covariance structure.

4.2 FFTW plan creation and reuse

FFTW separates plan creation (analyzing the transform size and selecting an algorithm) from execution [8]. The implementation creates two plans once before the Monte Carlo loop:

1. A forward *c2c* plan of length $M = 2N$ to compute the circulant eigenvalues from the first row c . This plan is executed once and immediately destroyed.
2. A backward *c2c* plan (FFTW_BACKWARD) of length $M = 2N$ to synthesise fGn increments per path. This plan is reused across all M_{paths} calls to `fftw_execute`.

I used full complex-to-complex (*c2c*) transforms rather than the real-input variants (`r2c/c2r`). The Hermitian-symmetry bookkeeping required for the latter, combined with the phase adjustments in the synthesis step, introduces enough notation to obscure the algorithm; *c2c* keeps the implementation transparent at negligible memory cost ($2N$ vs $N+1$ complex words for the eigenvalue array).

Both plans use `FFTW_ESTIMATE`, which skips empirical tuning. A production deployment could substitute `FFTW_MEASURE` or persist a wisdom file (§4.9) for repeated calls at the same N .

4.3 `std::complex<double>` and the FFTW ABI

A note for the interested reader / C++ developer: FFTW's C interface expects `fftw_complex*`, which is `double[2]`. I allocate `std::vector<std::complex<double>>` and pass a `reinterpret_cast<fftw_complex*>` of its data pointer. This cast is valid, as the C++11 standard mandates that `complex<double>` has the same memory layout as `double[2]`, with the real part first, making the two representations ABI-compatible. Using `std::vector` rather than a raw FFTW-allocated array lets RAII manage the buffer lifetime without a manual `fftw_free`.

4.4 FFT normalization convention

FFTW’s backward transform is unnormalized: the output of `fftw_execute` on a length- M array equals M times the true inverse DFT. To obtain fGn increments with the correct covariance structure, the per-component scaling factor is

$$s_j = \sqrt{\lambda_j/M}, \quad M = 2N, \quad (10)$$

not $\sqrt{\lambda_j}$. Applying the unnormalized backward transform to the scaled vector $w_j = s_j(a_j + ib_j)$ (with $a_j, b_j \stackrel{iid}{\sim} \mathcal{N}(0, 1)$) then yields $\text{Re}(\text{IFFT}(w))_{0:N}$ with covariance $\gamma(|i - j|)$, as required.

4.5 Randomized SVD: subspace iteration

The rSVD implementation (`rsvd.hpp`) draws a Gaussian test matrix $\Omega \in \mathbb{R}^{N \times (k+p)}$ with oversampling $p = 5$ and orthonormalizes the sketch, $Q = \text{orth}(C\Omega)$. It then applies $q = 2$ rounds of subspace iteration,

$$Q \leftarrow \text{orth}(C \text{orth}(C^\top Q)), \quad q = 2 \text{ times}, \quad (11)$$

where $\text{orth}(\cdot)$ is a thin Householder QR. This is Algorithm 4.4 of Halko et al. [13]. The QR after every product is what separates it from the plain power iteration of their Algorithm 4.3.

Re-orthonormalizing is necessary here, not cosmetic. Without it, the columns of $(CC^\top)^q C\Omega$ are scaled by singular values raised to the power $2q + 1 = 5$. At $N = 500$, $k = 128$ the ratio $(\sigma_1/\sigma_{k+p})^5$ is $\approx 2 \times 10^{15}$, at the limit of double precision, so the directions the sketch is meant to capture are at risk of being lost to roundoff. An earlier version of my code used Algorithm 4.3. Switching changed the Frobenius errors by less than 0.02% (relative) at the ranks tested, but it removes a failure mode at larger k or N . With subspace iteration, the Frobenius error at every rank is within about 2% (relative) of the optimal Eckart–Young truncation computed from the exact eigendecomposition. The slow error decay in §5.5 is therefore a property of C , not of the randomized algorithm.

4.6 In-place Cholesky and the triangular mat-vec

The per-path product νLz needs to touch only the $N(N+1)/2$ non-zero entries of L . My first implementation copied L out of `Eigen::LLT` into a dense `MatrixXd` and computed $L * z$. That is a general mat-vec which also multiplies the zero upper triangle: $2N^2$ flops instead of N^2 , and twice the bytes read per path. It also kept three $N \times N$ matrices alive during the Monte Carlo loop: C , the factorization’s internal copy, and L .

The current implementation factors in place, running `Eigen::LLT` on an `Eigen::Ref` to C so that L overwrites the lower triangle of C , and multiplies through a lower-triangular view of that storage (`triangularView<Lower>`). In isolation at $N = 1000$, the triangular mat-vec takes $\approx 62 \mu\text{s}$ against $\approx 115 \mu\text{s}$ for

the dense one. End to end, the Cholesky run at $N = 1000$, $M = 10,000$ drops from 1.61 s to 1.13 s, and resident memory drops from 3×7.6 MB to 7.6 MB. Prices are unchanged to six significant figures, since the random draws and the arithmetic on the non-zero entries are the same. This correction matters for the comparison in §5.1: against the dense baseline the FFT method looked $1.5\times$ faster at $N = 1000$, but against the fair one it is only $\approx 9\%$ faster.

4.7 Piecewise-constant volatility convention

All three pricers share a single GBM step kernel (`asian_payoff.hpp`):

$$S_i = S_{i-1} \exp\left((r - \tfrac{1}{2}\sigma_i^2) \Delta t + \sigma_i \sqrt{\Delta t} Z_i\right), \quad \sigma_i = e^{\log \sigma[i]}, \quad (12)$$

where $\log \sigma[i] = \nu W_{t_i}^H$ is the fBM log-vol evaluated at the end of the i -th interval (right-endpoint convention), and $Z_i \stackrel{iid}{\sim} \mathcal{N}(0,1)$ is an independent Brownian increment for the price process. This departs from the standard Euler-Maruyama left-endpoint rule, which would use σ_{i-1} (the volatility at the start of the interval). The departure does not constitute a look-ahead bias in the causal sense: W^H is pre-generated independently of the price process S , so using σ_i does not introduce any dependence on future values of S . It does introduce an $O(\Delta t)$ discretization error relative to the left-endpoint rule, which at $N = 252$ is negligible. Using the same convention across all three pricers ensures that timing and price comparisons are not confounded by volatility discretization differences.

4.8 Header-only pricers and ODR

Each algorithm lives entirely in a `.hpp`: `cholesky.hpp`, `fft.hpp`, and `lowrank.hpp`. Each `price()` function is marked `inline`:

```
namespace cholesky { inline double price(...) { ... } }
```

`inline` permits the same function to be defined in multiple translation units; the linker deduplicates the definitions and keeps one copy in the final binary. This allows `benchmark.cpp` to include all three headers in one translation unit without ODR violations, while each `_pricer.cpp` standalone executable also includes its header and calls `price()` directly.

4.9 Optimization opportunities not implemented

Three straightforward improvements would accelerate the MC loop without algorithmic changes:

Two fGn paths per inverse FFT At $N = 1000$ the FFT pricer’s per-path cost is dominated not by the transform but by random number generation. In isolation, its $4N$ Gaussian draws (real and imaginary parts of $2N$ complex variates) take $\approx 60\text{--}70\ \mu\text{s}$, against $\approx 8\ \mu\text{s}$ for the size- $2N$ inverse FFT. The real and imaginary parts of the transformed vector each have exactly the fGn

Table 2: Fitted power-law exponents α from log-log regression $\log t = \log c + \alpha \log N$ over $N \in \{252, 500, 1000\}$. Theoretical exponent assumes $O(MN^\alpha)$ total work.

Method	Fitted α	Fitted c	Theoretical α
Dense Cholesky	1.33	1.1×10^{-4}	2.0
Circulant + FFT	0.95	1.5×10^{-3}	1.0
Low-rank rSVD ($k = 32$)	1.02	3.6×10^{-4}	1.0

covariance, and their cross-covariance vanishes because $\lambda_j = \lambda_{2N-j}$; I confirmed both numerically to within Monte Carlo tolerance. The pricer nevertheless discards the imaginary part. Using both would halve the Gaussian draws per fBM path, which is the largest constant-factor saving available to Algorithm 2.

OpenMP path-level parallelism The outer `for(m)` loop over paths is embarrassingly parallel: each path reads only the shared, read-only factor L (or L_k , or the FFTW eigenvalues) and accumulates independently. A `#pragma omp parallel for reduction(+:payoff_sum)` decorator with thread-local RNG state would give near-linear multi-core scaling with no algorithmic changes.

FFTW wisdom files The current code uses `FFTW_ESTIMATE`, which accepts the default transform plan. For repeated calls at a fixed N (e.g., in a calibration loop that prices the same contract many times), `FFTW_MEASURE` or a pre-saved wisdom file would eliminate the one-time planning overhead by caching the optimal decomposition into FFT subproblems [8].

5 Algorithm Benchmarks

All C++ benchmarks were run single-threaded on an Apple M2 (8-core, 24 GB unified memory, rated DRAM bandwidth 100 GB/s; its largest cache is the 16 MB L2 shared by the performance cores, and it has no L3), compiled with `-O3 -march=native -DNDEBUG`, with $M = 10,000$ paths and $N \in \{252, 500, 1000\}$. Timings come from a single benchmark run; back-to-back repeats of the Cholesky pricer varied by up to $\approx 15\%$. Reference prices use $M = 500,000$ paths (see §5.5).

5.1 Runtime scaling

Measurement setup: controlled: $M = 10,000$ paths; independent: $N \in \{252, 500, 1000\}$ and method; dependent: total wall-clock time.

Figure 2 shows the results. Log-log regression over three N values gives the fitted exponents in Table 2.

The Cholesky empirical exponent ($\alpha \approx 1.33$) is substantially below the theoretical 2.0. This occurs because the per-path cost is not purely the $O(N^2)$

triangular multiply: drawing N Gaussian variates, computing N exponentials for the stock price path, and accumulating the payoff each contribute $O(N)$ overhead that is non-negligible for $N \leq 1000$. As N grows further, the N^2 term will eventually dominate and the exponent will approach 2.0; the tested range is still in the crossover region.

The FFT and rSVD exponents are both ≈ 1.0 , consistent with theory. The $O(\log N)$ factor distinguishing $O(N \log N)$ from $O(N)$ is unresolvable over a $4\times$ range in N (a log-ratio of $\log(1000)/\log(252) \approx 1.25$), and appears as noise in the regression.

Absolute times matter as much as exponents. At $N = 1000$ the three methods take 1.13 s (Cholesky), 1.04 s (FFT), and 0.43 s (rSVD, $k = 32$). The FFT method is only $\approx 9\%$ faster than Cholesky, even though its sampling step does an order of magnitude less arithmetic per path ($\approx 10^5$ flops for a size- $2N$ complex FFT against $N^2 = 10^6$ for the triangular mat-vec). Two effects explain the gap. First, the price-path work (N Gaussian draws, $2N$ exponentials, and the payoff average) is identical for all three methods and is a large share of every per-path cost. Second, the FFT sampler draws $4N$ Gaussian variates per path against N for Cholesky, and at $N = 1000$ those draws cost about eight times as much as the inverse FFT itself (§4.9). The asymptotic advantage is real, and the gap widens with N ($\alpha = 1.33$ against 0.95), but at $N \leq 1000$ it is largely hidden by work the two methods share. With the earlier dense- L baseline (§4.6) the FFT method appeared $1.5\times$ faster, a reminder that a speedup is only as meaningful as its baseline.

5.2 Memory and cache pressure

Controlled: $M = 10,000$ paths; Independent: N , method; Dependent: size of the dominant arrays resident during the MC loop (C++), peak heap allocation (Python via `tracemalloc`). For C++ I report the analytical size rather than measured RSS. The benchmark’s RSS change across a call is not a peak, because buffers are freed before the call returns, and it can read zero.

At the tested sizes the Cholesky loop is not DRAM-bound. At $N = 1000$, L occupies $N^2 \times 8 = 7.6$ MB (the full matrix is allocated, though only its lower half is read), which fits in the 16 MB L2. Dividing the bytes of L read in a run by its wall time gives an effective streaming rate of 14, 24, and 35 GB/s at $N = 252, 500$, and 1000, well below the 100 GB/s DRAM rating. Because L is cache-resident, these figures measure how fast the loop consumes L , not DRAM traffic. Beyond $N \approx 1450$, L no longer fits in L2 and every path must stream $4N(N+1)$ bytes from DRAM, at which point the loop would become bandwidth-limited [7].

The rSVD method has two memory variants. The standard variant retains C in memory during the MC loop (7.6 MB at $N = 1000$, the same footprint as Cholesky). A second variant, `price_freed_timed()`, releases C immediately after forming L_k ; the resident footprint during the MC loop is then $O(Nk)$ (0.24 MB at $N = 1000$, $k = 32$), far below the L2 capacity. The FFT method needs only four length- $2N$ complex buffers, 128 KB at $N = 1000$.

The Python RFSV engine (used in Sections 6 and 6.3) is vectorized across paths and so has a different profile. `tracemalloc` on the log-volatility generator measures a peak of 104 bytes per (path, time step), i.e. 13 float64 words per element of the (M, N) output, at every $M \geq 1000$. That count is consistent with the working arrays the generator holds at once, which are $2N$ wide: two real Gaussian arrays and two complex arrays of shape $(M, 2N)$ already account for 12 words per output element. At $M = 10,000$, $N = 252$ the peak is ≈ 260 MB, far beyond any cache. The measurement covers path generation only; the price-path arrays come on top.

5.3 MC convergence

Controlled: $H = 0.10$, $\nu = 0.30$, $K = 100$, $T = 1$, $S_0 = 100$, $r = 0$, $N = 252$; Independent: $M \in \{100, 250, 500, 1000, 2500, 5000, 10000, 25000\}$; Dependent: mean price and cross-seed standard deviation over 5 independent random seeds.

Figure 6 shows the results. The fitted log-log slope is -0.711 ($R^2 = 0.959$), steeper than the theoretical -0.5 from the central limit theorem [10]. The discrepancy arises from the small- M regime: with only 5 seeds, the cross-seed standard deviation at $M \leq 500$ is a noisy chi-squared estimate (4 degrees of freedom) with high variance, pulling the regression slope away from -0.5 . (The sample standard deviation is in fact a downward-biased estimator of σ at finite sample sizes by Jensen’s inequality; the dominant effect here is its large variability, not a directional bias.) The theoretical line at slope -0.5 is shown as a dotted reference in Figure 6(b); the data approach it at large M but deviate below $M \approx 1,000$. The cross-seed standard deviation at small M is also noisy because the payoff itself is highly variable, with $\sigma_{\text{payoff}} \approx 61$ (the sample standard deviation over 200,000 paths). An earlier version of this study estimated $\sigma_{\text{payoff}} \approx 35$ by rescaling the spread of only five seed-level prices, an estimate with roughly 35% relative error; the current code uses the per-path payoffs directly. The payoff is so variable because our RFSV model uses $\sigma_0 = e^{\nu \cdot 0} = 1.0$ (100% annualised base volatility), producing highly variable payoffs. This is not a model defect but a consequence of using the uncalibrated log-vol parameterisation $\log \sigma_t = \nu W_t^H$; the IV validation in §6.1 auto-calibrates a drift $\mu_0 = \log \sigma_{\text{target}}$ to match market volatility levels. A calibrated $\mu_0 \approx \log(0.18) \approx -1.7$ would reduce σ_{payoff} by a factor of ~ 5 and the path count required for the same precision by ~ 25 ; the uncalibrated parameterisation was retained to keep the convergence study independent of the IV calibration procedure.

To achieve price uncertainty below 0.5 units (roughly 2% of the ATM price ≈ 23.6) at 95% confidence requires $M \approx (2\sigma_{\text{payoff}}/0.5)^2 \approx 60,000$ paths. I use $M = 10,000$ for most experiments, accepting $\approx \pm 1.2$ units per run ($2 \times \text{SE}$, the 95% confidence half-width). Where differences between runs matter, I use common random numbers or means over multiple seeds. One caveat: the reference line in Figure 6 is the C++ reference at $N = 500$, while this study simulates at $N = 252$; the averaging grids differ, so the line is a visual guide rather than the exact target.

5.4 Numerical stability

(a) FFT eigenvalue positivity: Figure 7(a) sweeps $H \in [0.05, 0.501]$ at $N = 252$ and $N = 1008$ and plots the smallest circulant eigenvalue relative to $\gamma(0)$. No eigenvalue is negative at any H or N tested, so nothing is clipped and the FFT sampler is exact. The minima agree with the closed form (8) to every printed digit: 0.24% at $H = 0.10$, $N = 252$ and 0.08% at $N = 1000$, always attained at λ_0 . The margin collapses as $H \rightarrow 0$, falling to 0.07% at $H = 0.05$, $N = 252$, but it is positive by construction and many orders of magnitude above double-precision roundoff. At $H = 1/2$ the increments are white noise, every eigenvalue equals $\gamma(0)$, and the ratio is 1. Consistent with exactness, the 500,000-path FFT and Cholesky prices in §5.5 agree to within MC noise.

(b) rSVD condition number: Figure 7(b) plots $\kappa(L_k) = \sqrt{S_{\max}}/\sqrt{S_{\min}}$ as a function of rank k at $H = 0.10$, $N = 252$. The condition number grows from 8.2 at $k = 8$ to 23.3 at $k = 64$: as we include smaller singular values (the denominator shrinks), κ grows. A condition number of 23 is well within the range of float64 arithmetic ($\epsilon \approx 10^{-16}$, so $\kappa \cdot \epsilon \approx 2 \times 10^{-15}$) and does not cause numerical issues in path generation.

(c) Cholesky covariance conditioning: Figure 7(c) shows $\kappa(C) = \lambda_{\max}/\lambda_{\min}$ vs. N . At $N = 252$, $\kappa(C) \approx 789$; at $N = 1000$, $\kappa(C) \approx 4 \times 10^3$. The fitted power law is $\kappa \approx 1.08 N^{1.19}$: the covariance matrix becomes increasingly ill-conditioned as more time steps are added, but only polynomially. This growth is expected: adding time steps increases correlations between adjacent points, pushing $\lambda_{\min}$ toward zero. The matrix remains positive-definite and Cholesky-factorizable throughout the tested range. **Eigen::LLT** fails when floating-point roundoff destroys positive-definiteness, which occurs when $\kappa(C) \cdot \epsilon_{\text{mach}} \geq 1$, i.e., $\kappa(C) \geq 10^{16}$. At $N = 1000$ where $\kappa(C) \approx 4 \times 10^3$, this threshold is twelve orders of magnitude away; the fit would not reach it until $N \sim 10^{12}$.

The slow singular value decay visible in Figure 1(d) is a related but distinct phenomenon: a large $\kappa(C)$ means the spectrum spans many decades, but slow decay (many non-negligible eigenvalues) is what actually hinders low-rank approximation. A matrix can have an enormous condition number and still be perfectly approximated at low rank if the eigenvalues drop off rapidly after the first few; in the fBM case, rough H implies both a large condition number and a slow decay rate. As Figure 1(d) shows, that slow decay belongs to the full matrix; its well-separated off-diagonal blocks still compress to rank ≤ 3 .

5.5 Low-rank approximation quality

Controlled: $N = 500$, $M = 10,000$ paths (rSVD) and $M = 500,000$ paths (reference); Independent: rSVD target rank $k \in \{2, 4, 8, 16, 32, 64, 128\}$; Dependent: Frobenius error $\|C - C_k\|_F/\|C\|_F$ and relative price error $|\hat{p}_k - p_{\text{ref}}|/p_{\text{ref}}$.

Figure 8 shows results. The reference price $p_{\text{ref}} = 23.58$ is the average of a 500,000-path Cholesky run (23.62) and a 500,000-path FFT run (23.54);

averaging two independent exact methods avoids privileging either as ground truth. With $\sigma_{\text{payoff}} \approx 61$, each run has a standard error of ≈ 0.09 . Their difference of 0.08 is below the 0.12 standard error of a difference, consistent with both methods being exact, and the pooled reference has a standard error of ≈ 0.06 .

Two accuracy metrics are reported because they measure different things. The Frobenius error $\|C - C_k\|_F / \|C\|_F$ measures matrix approximation quality exactly and without noise. It decreases monotonically but slowly: 8.5% at $k = 2$, 3.5% at $k = 8$, 1.9% at $k = 32$, and 1.2% at $k = 128$. The price error measures end-to-end accuracy but carries MC noise of ≈ 0.6 units at $M = 10,000$, i.e. $\approx 2.6\%$ of p_{ref} .

The key limitation of this experiment is that the price errors are not resolved. The observed relative errors range from 0.2% ($k = 4$) to 3.1% ($k = 2$), and every one lies within ≈ 1.2 standard errors of zero, so they do not distinguish one rank from another. The non-monotone pattern (2.6% at $k = 64$ against 0.2% at $k = 4$) is noise. Resolving a price bias of 0.1 units would need on the order of $M = 10^6$ paths per rank. The Frobenius error is therefore the meaningful accuracy metric here. Its slow decay matches Figure 1(d): the spectrum of the full matrix C decays only algebraically, because the rough kernel’s singularity lies on the diagonal, where a global rank- k approximation cannot isolate it.

6 Model Validation

Three experiments probe the RFSV pricer: (i) the implied volatility smile against live SPY option chain data; (ii) a comparison with the Lévy (1992) analytical approximation and the roughness premium; and (iii) a parameter sensitivity sweep over H and ν . Each uses the Python RFSV engine (`data/rfsv_model.py`), which implements the circulant-FFT path generator with the same normalization as the C++ FFT pricer.

6.1 Implied volatility smile

Figure 9 shows the results. Because the uncalibrated RFSV model sets $\sigma_0 = e^{\nu \cdot 0} = 1.0$ (100% annualised vol) while SPY trades near $\sigma \approx 14\%$ at the money, a direct level comparison is not meaningful. I instead add a log-vol drift: $\log \sigma_t = \mu_0 + \nu W_t^H$, calibrating $\mu_0 = \log \sigma_{\text{eff}}$ per expiration to match the RFSV ATM price to the market ATM price. This aligns the curves at the money (13.9% and 14.5% for the two expirations) so the comparison concerns smile shape (slope and curvature), not the vol level.

The comparison is informative mainly about what the model, as implemented, cannot do. The RFSV smile is shallow and has its minimum at the money: for the nearer expiration it rises from 13.9% at the money to $\approx 14.3\%$ at $K/S = 0.965$ and $\approx 14.7\%$ at $K/S = 1.05$. The SPY smile is strongly skewed instead. It falls from 21.7% at $K/S = 0.965$ to a minimum of $\approx 10.7\%$ near $K/S = 1.03$ and recovers only to 13.4% at $K/S = 1.05$; the later expiration shows the same

shape. This is expected. The engine draws the price shocks independently of the volatility driver, so the spot-vol correlation is $\rho = 0$, and with $\rho = 0$ a stochastic volatility model produces a smile that is symmetric in log-moneyness. It can generate curvature but not skew. The steep short-dated skew that rough volatility is known for [9] requires $\rho < 0$ (§7.4).

A Monte Carlo pitfall: forward error masquerading as skew. An earlier run of this experiment used $M = 3,000$ paths with common random numbers but no correction, and its RFSV curve rose steadily with strike, from 13.0% to 15.7% across $0.98 \leq K/S \leq 1.05$. That curve looks like an upward skew, which a $\rho = 0$ model cannot produce. The cause was sampling error in the forward. The sample mean of S_T was 0.048% below S_0 , and because every strike shares the same paths, that error acts like a shifted forward and tilts the entire IV curve; it also biases the calibrated μ_0 . A 400,000-path run of the same model gives an approximately symmetric smile with its minimum at the money (14.57% at $K/S = 1.00$, against 14.79% at 0.98 and 14.70% at 1.02). The current script rescales the simulated S_T so that its sample mean equals the forward exactly, a standard moment-matching correction [10], and uses $M = 20,000$ paths. At $N \approx 5$ time steps per path the whole experiment runs in a few seconds.

Limitations I set $r = 0$ and ignore SPY’s dividend yield. For near-the-money options this shifts IV levels slightly and leaves the shape comparison largely unaffected. For deep in-the-money calls the effect is large: the carry in an ITM call price is misread as time value, which inflates those market IVs (many above 40%), so I do not interpret the low-strike wing. SPY options are American-style; for calls on a low-dividend ETF the early-exercise premium is small, and I treat them as European. The comparison is a single-day snapshot; market IV smiles vary substantially day-to-day. Brent’s method stops at a tolerance of 0.01 in μ_0 , about 1% in σ_{eff} .

6.2 Lévy benchmark and roughness premium

Figure 10 shows the results.

Lévy is a weak benchmark at 100% volatility: When $\nu \rightarrow 0$, $\sigma_t = e^{\nu W_t^H} \rightarrow 1$ pointwise, so the RFSV dynamics reduce to constant-volatility GBM with $\sigma = 1$, and the Lévy approximation is the natural closed-form comparison [15]. The approximation is designed for typical equity volatilities, however, and at $\sigma = 1$ it is visibly biased. At $K = 100$ it gives 23.72, while exact discrete GBM by Monte Carlo (200,000 paths) gives 22.29 ± 0.12 : Lévy overprices by $\approx 6\%$. This explains panel (a), where the dashed Lévy curve lies on or above all four RFSV curves at $K \leq 100$ and at $K = 120$, even though stochastic volatility should raise prices above GBM. The Lévy curve therefore cannot serve as a sanity check on the engine; a $\nu = 0$ Monte Carlo run is the correct GBM baseline. Against that baseline the direction is as expected: the RFSV reference price

$p_{\text{ref}} = 23.58$ (§5.5) is ≈ 1.3 units above the GBM price, though on a different time grid ($N = 500$ versus $N = 252$).

Roughness premium: The roughness premium in panel (b) is $\hat{p}(H=0.10) - \hat{p}(H=0.50)$, both at $\nu = 0.30$ and $\mu_0 = 0$ (uncalibrated, $\sigma_0 = 1.0$), averaged over 3 seeds of $M = 10,000$ paths. The premium is positive at all nine strikes, between 0.42 and 0.91 price units, with the largest value at $K = 105$. The ATM value of 0.55 is consistent with the sensitivity sweep of §6.3, which gives 0.44 at the same (H, ν) .

Limitations: The roughness premium as defined here conflates roughness with total variance. At fixed ν , $\text{Var}(\log \sigma_t) = \nu^2 t^{2H}$ is larger for small H when $t < 1$, and the expected integrated variance $\mathbb{E}[\int_0^T \sigma_t^2 dt] = \int_0^T e^{2\nu^2 t^{2H}} dt$ is ≈ 1.16 at $H = 0.10$ against ≈ 1.10 at $H = 0.50$. A 6% increase in variance raises an at-the-money price by roughly 3%, since the price scales approximately with volatility; here that is ≈ 0.7 units, comparable to the entire measured premium. The experiment therefore does not separate roughness from variance. A variance-normalised comparison, adjusting $\nu(H) = \nu_0 \sqrt{\frac{2H+1}{2}} T^{1-2H}$ so that $\int_0^T \text{Var}(\log \sigma_t) dt$ is the same for every H , would isolate the roughness effect; it is proposed in §7.3.

On noise: each price in panel (a) is a 3-seed mean with a standard error of $\approx 61/\sqrt{30,000} \approx 0.35$. The premium is a difference of two such means, but because the same seeds are used for every H the two share their random numbers, and the difference is far less noisy than the ≈ 0.5 that independent estimates would give. I did not measure its standard error directly; its sign is consistent across all nine strikes.

6.3 Parameter sensitivity:

Figures 11 and 12 show the results:

- **Price decreases with H :** at $\nu = 0.30$ the ATM price falls from 23.60 at $H = 0.10$ to 23.17 at $H = 0.50$, and the effect grows with ν (0.80 units at $\nu = 0.40$, under 0.03 at $\nu = 0.10$). Lower H at fixed ν raises $\text{Var}(\log \sigma_t)$ as well as roughness (§6.2), and a more variable σ_t widens the distribution of the average A , which raises the expected value of the convex payoff $\max(A - K, 0)$. The absolute spread across H widens toward out-of-the-money strikes (Figure 12).
- **Price increases with ν :** at $H = 0.10$ the ATM price rises from 22.74 at $\nu = 0.10$ to 24.31 at $\nu = 0.40$. Larger vol-of-vol amplifies σ_t variation, and the increase is convex in ν because $\sigma_t = e^{\nu W_t^H}$ enters exponentially.
- **Price decreases with strike K :** standard convexity of the call payoff.

At the literature parameters $H = 0.10$, $\nu = 0.30$ with the uncalibrated base volatility $\sigma_0 = e^{\nu \cdot 0} = 1.0$ (100% annualised), the ATM price is 23.60, consistent with the reference price $p_{\text{ref}} = 23.58$ from §5.5 to well within MC error. The 100% base vol level is not calibrated to market prices here (see §6.1 for that step). Moving to the Brownian limit $H = 0.50$ at the same ν gives 23.17, a premium of 0.44 price units ($\approx 1.9\%$). An earlier draft of this report quoted a premium of 2.1 units; that run used an independent seed for every cell, and its cell-to-cell differences were dominated by MC noise.

Limitations: All grid points share one random seed. This makes the trends across the grid reliable, but the whole surface carries a common MC offset with a standard error of ≈ 0.6 price units, so absolute prices are uncertain at that level. The sensitivity surfaces are computed on a 6×4 grid ($H \times \nu$) at a fixed $N = 252$; finer grids and higher M are feasible with the C++ benchmark but were not run for this report.

7 Future work

The three algorithms implemented here exploit three structural properties of the fBM covariance matrix: positive-definiteness (Cholesky), stationarity of increments (circulant FFT), and smooth off-diagonal low-rank structure (global rSVD).

7.1 True hierarchical H-matrix

The global rSVD (Algorithm 3) approximates the entire $N \times N$ matrix C at rank k . This forces the rank budget to cover both the smooth far-off-diagonal region and the singular near-diagonal region, where $C(t, s) \sim |t - s|^{2H}$ with $H = 0.10$ is Hölder-0.2 continuous and barely above C^0 regularity. The global approximation cannot specialize, which is why the spectrum of the full matrix decays slowly at small H even though its off-diagonal block compresses to rank ≤ 3 (Figure 1(d)).

A true hierarchical H-matrix separates these concerns by a recursive block-tree decomposition of C [12, 2]. At each level of the tree:

- Near-diagonal blocks (time intervals that overlap or touch) are stored dense. The near-diagonal singularity is not low-rank, so no rank budget is wasted trying to compress it.
- Far-off-diagonal blocks (well-separated time intervals) lie entirely in the smooth region $t \neq s$ and are approximated at low rank proportional to the distance from the diagonal.

The result is an H-Cholesky factorization with $O(Nk \log N)$ construction cost and $O(Nk \log N)$ per-path cost (H-matrix triangular solve traverses the block tree), versus the global rSVD's $O(Nk)$ per-path cost. The H-matrix advantage

lies in construction: $O(Nk \log N)$ vs $O(N^2k)$ for the global rSVD sketch, a factor of $N/\log N$ for $k \ll N$.

For the rough case $H = 0.10$, the off-diagonal block needs rank 3 rather than 2 for 1% accuracy (Figure 1(d)): slightly more than at $H = 0.50$, but far below the ranks the global approximation uses. The H-matrix benefit over the global rSVD grows as N increases beyond the current test range. At $N = 252$, $O(N^2k) = O(252^2 \cdot 64) \approx 4 \times 10^6$ flops for construction; the H-matrix would reduce this by roughly $\log_2 252 \approx 8$ for equal per-block rank, which becomes worthwhile at $N \geq 1000$.

7.2 Hybrid Scheme for rough Bergomi

The Hybrid Scheme of Bennedsen et al. [3] achieves more than 80% RMSE reduction over naïve Euler discretization at $O(N \log N)$ cost by exploiting the separability of the Volterra kernel $K(t, s) = (t - s)^{H-1/2}$ near $t = s$: the singular part is handled analytically and the smooth remainder is convolved via FFT. This is a structurally superior simulation method for the rough Bergomi model of Bayer et al. [1], in which log-volatility is a Volterra integral $\int_0^t (t - s)^{H-1/2} dW_s$.

The RFSV model studied here uses a different representation: $\log \sigma_t = \nu W_t^H$, where W_t^H is a standard fBM (not a Volterra integral with the Bergomi kernel). The two models are different processes with different correlation structures, though both have Hurst exponent H controlling roughness. Adopting the Hybrid Scheme would require switching to the rough Bergomi model class, which has distinct option pricing properties and calibration procedures. The structural insight of separating the kernel into a singular analytic part and a smooth FFT-able remainder is directly relevant to the fBM kernel $C(t, s)$ and motivates the hierarchical treatment in §7.1.

7.3 Variance-normalised roughness premium

The roughness premium in §6.2 is computed as $\hat{p}(H=0.10) - \hat{p}(H=0.50)$ at fixed $\nu = 0.30$. At fixed ν , the total integrated variance

$$\mathbb{E} \left[\int_0^T \sigma_t^2 dt \right] = \int_0^T e^{2\nu^2 t^{2H}} dt \quad (13)$$

depends on H : rougher paths (smaller H) have larger integrated variance because $t^{2H} > t$ on $(0, 1)$ when $H < 1/2$. The premium therefore conflates two effects: path regularity and total variance.

A variance-normalised comparison holds integrated variance constant by adjusting ν as a function of H . To leading order, matching the short-time variance gives

$$\nu(H) = \nu_0 \sqrt{\frac{2H+1}{2} T^{1-2H}}, \quad (14)$$

with ν_0 fixed at the Brownian reference ($H = 1/2$). Running §6.2 with $\nu(H)$ rather than fixed ν would isolate the pure roughness effect, resolving the confound noted in §6.2.

7.4 Spot-volatility correlation

The engine draws the price shocks Z independently of the fBM driving volatility, so the spot-vol correlation is $\rho = 0$. As §6.1 shows, this leaves the model unable to produce the downward skew of equity smiles. Introducing $\rho \neq 0$ requires simulating W^H jointly with the Brownian motion that drives the price. The dense Cholesky sampler can absorb this directly by factoring the $2N \times 2N$ joint covariance, which is explicit when W^H is written as a Volterra integral of that Brownian motion; the circulant embedding would need to be extended to the joint process. This is the single change most needed before the IV comparison can test the model rather than its simplifications.

7.5 Quasi-Monte Carlo path generation

All three algorithms use pseudo-random Gaussian draws, giving a standard MC error of $O(M^{-1/2})$. Replacing these with low-discrepancy sequences [17] reduces the effective error rate to approximately $O(M^{-1}(\log M)^N)$, close to $O(M^{-1})$ in practice [10]. This improvement is orthogonal to the choice of covariance simulation method: any of the three algorithms can be combined with QMC by replacing the standard normal draws.

In $N = 252$ dimensions the low-discrepancy property is strongest for the first few coordinates. Brownian bridge dimension ordering concentrates the largest variance components in the lowest-index dimensions, ensuring the low-discrepancy sequence acts on the high-information components first [14]. At $M = 10,000$, the MC standard error of ≈ 0.6 (§5.3) would fall to ≈ 0.006 under ideal $O(M^{-1})$ QMC, a $100\times$ reduction at no additional computational cost, improving the statistical resolution of the roughness premium and parameter sensitivity results.

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Reproducibility

All source code, benchmark scripts, and plotting code are available at <https://github.com/t-flora/fbm-fast-pricers>. Results can be reproduced end-to-end by running `./run_pipeline.sh` from the project root. A manifest recording the git commit, parameters, and wall time is written to `plots/runs/TIMESTAMP/manifest.txt` for each run.

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Structural analysis: why FFT and low-rank rSVD work
 $H = 0.1$, $N = 64$ (heatmaps), $N = 128$ (SVD)

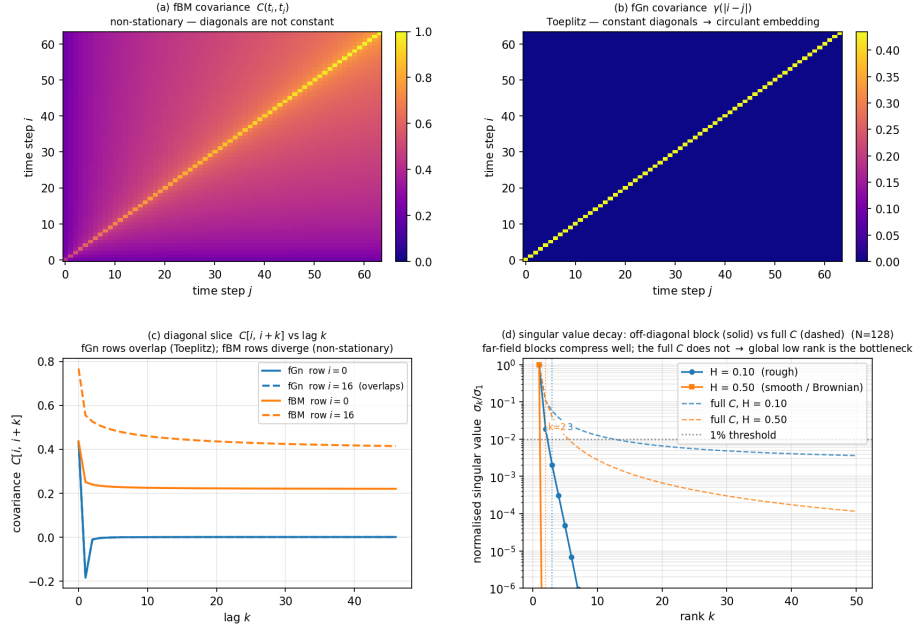


Figure 1: Structural properties of the fBM and fGn covariance matrices ($H = 0.10$). **(a)** fBM covariance $C(t_m, t_n)$: dense, not Toeplitz (variance grows with time). **(b)** fGn covariance $\gamma(|m - n|)$: Toeplitz (constant diagonals). **(c)** Diagonal slice $C[m, m + k]$ vs. lag k : fGn rows for $m = 0$ and $m = N/4$ overlap (Toeplitz); fBM rows diverge (non-stationary). **(d)** Normalised singular values σ_k/σ_1 of the off-diagonal block (solid) and normalised eigenvalues of the full matrix C (dashed) at $H = 0.10$ and $H = 0.50$. The off-diagonal block compresses to rank ≤ 3 at both H ; the full spectrum decays slowly, most slowly at $H = 0.10$. Controlled: $H = 0.10$, $N = 64$ (heatmaps), $N = 128$ (SVD). Independent: process type or Hurst exponent. Dependent: covariance pattern and singular value magnitude.

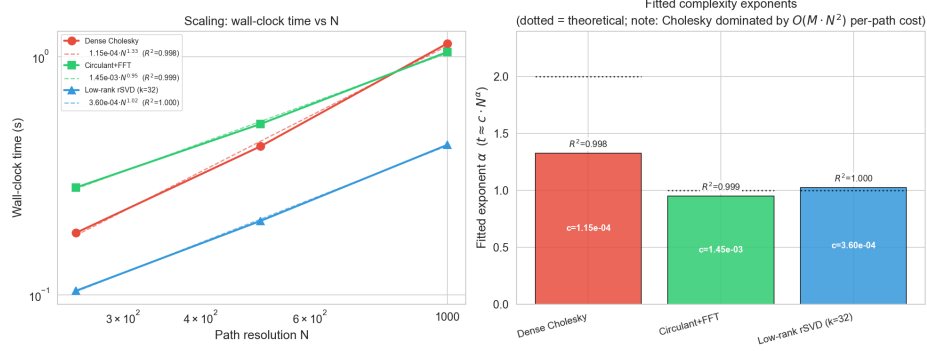


Figure 2: Wall-clock time vs. N on log-log axes (left panel) with fitted power laws $t = c \cdot N^\alpha$, and fitted exponents α with prefactors c (right panel; dotted line = theoretical exponent). Controlled: $M = 10,000$ paths. Independent: $N \in \{252, 500, 1000\}$, method. Dependent: total wall-clock time.

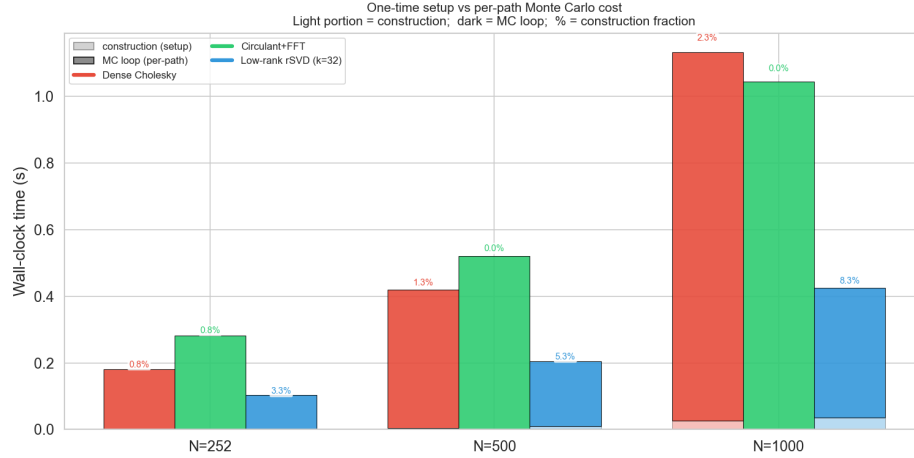


Figure 3: One-time setup vs. per-path Monte Carlo cost breakdown. Light portion = construction (factorization or eigenvalue computation); dark portion = MC loop. Percentages label the construction fraction. The FFT setup is negligible ($< 1\%$); rSVD setup grows with N because forming L_k requires a full $O(N^2k)$ sketch; Cholesky setup is a fixed $O(N^3)$ fraction that is small relative to the $O(MN^2)$ MC cost at $M = 10,000$. Controlled: $M = 10,000$ paths. Independent: $N \in \{252, 500, 1000\}$, method. Dependent: wall-clock time split between construction and MC loop.

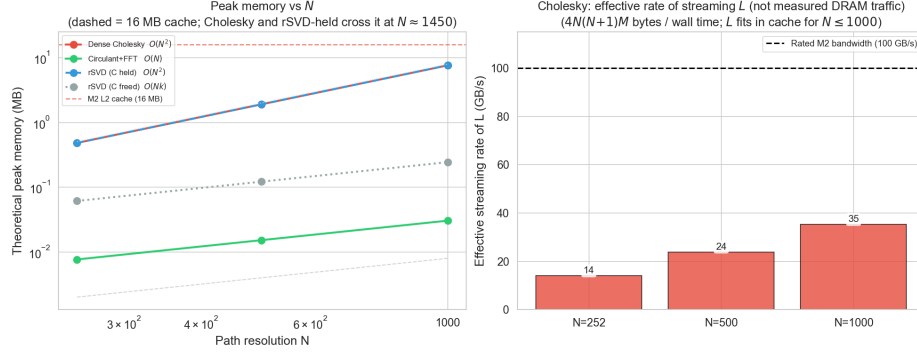


Figure 4: Size of the dominant array held during the MC loop vs. N for all C++ methods (left), and the effective rate at which the Cholesky loop streams L (right). The dashed horizontal line marks the M2’s 16 MB L2 cache; Cholesky and the rSVD variant that retains C cross it at $N \approx 1450$. The rSVD variant that frees C before the MC loop stays far below it. Controlled: $M = 10,000$ paths. Independent: N , method. Dependent: memory (MB) and effective streaming rate (GB/s), computed as bytes of L read divided by wall time; this is not a DRAM measurement.

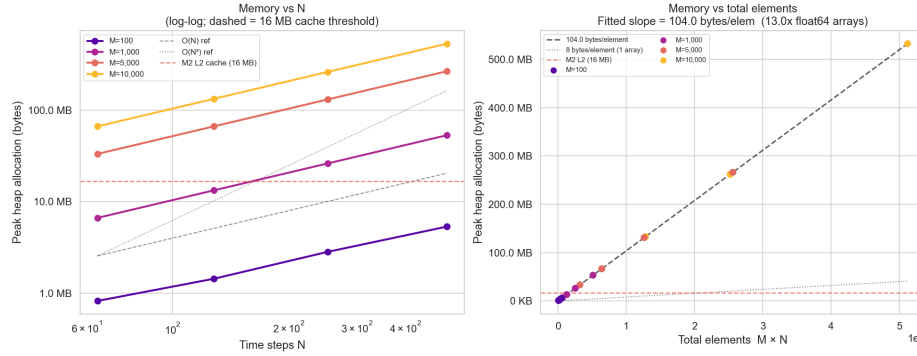


Figure 5: Python RFSV engine peak heap allocation measured by `tracemalloc`. (a) Peak bytes vs. N (log-log) for four path counts M ; the dashed red line marks the M2’s 16 MB L2 cache. Scaling is $O(MN)$ (parallel to the $O(N)$ reference), confirming the dominant allocation grows linearly with both M and N . (b) Peak bytes vs. total elements $M \times N$; the fitted slope ≈ 104 bytes/element is 13 float64 words per output element, consistent with the $(M, 2N)$ real and complex working arrays of the path generator. Controlled: $H = 0.10$, $\nu = 0.30$. Independent: $N \in \{64, 128, 252, 512\}$, $M \in \{100, 1000, 5000, 10000\}$. Dependent: peak heap allocation (bytes).

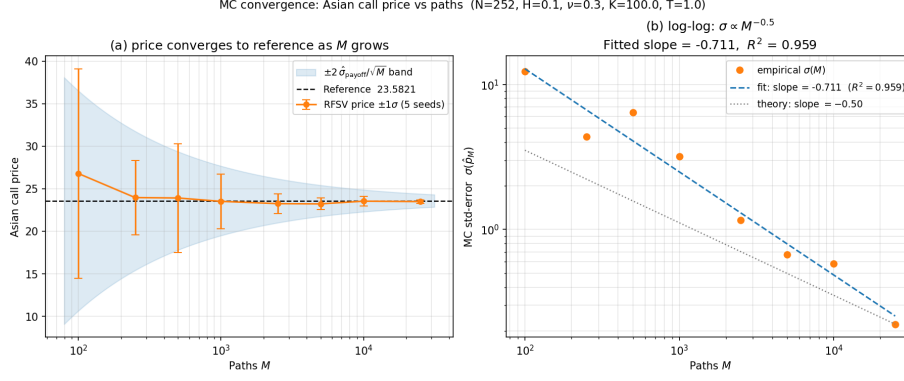


Figure 6: MC convergence: price $\pm 1\sigma$ vs. number of paths M (left), and log-log plot of cross-seed standard deviation vs. M with fitted slope (right). The shaded band in the left panel is $\pm 2\hat{\sigma}_{\text{payoff}}/\sqrt{M}$ around the C++ reference price, with $\hat{\sigma}_{\text{payoff}}$ the sample standard deviation of the 25,000 individual payoffs of one run. Controlled: $N = 252, H = 0.10, \nu = 0.30$. Independent: M (left and right); 5 seeds (right). Dependent: mean price $\pm 1\sigma$ and log-log slope.

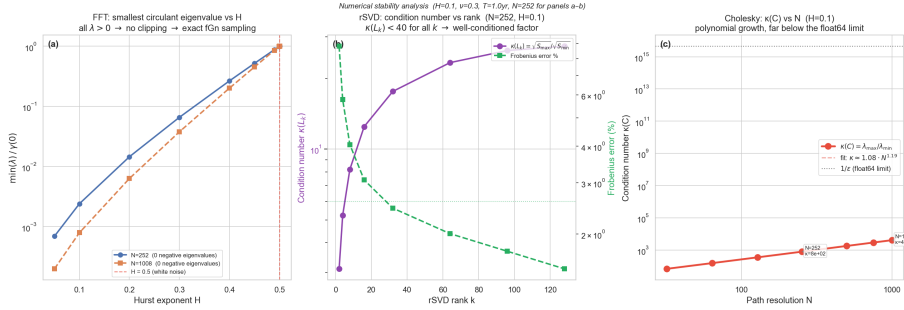


Figure 7: Numerical stability of the three methods. **(a)** FFT: smallest circulant eigenvalue relative to $\gamma(0)$ vs. Hurst exponent H ($N = 252$ and $N = 1008$); no eigenvalue is negative anywhere. **(b)** rSVD: condition number $\kappa(L_k)$ and Frobenius error vs. rank k ($N = 252, H = 0.10$). **(c)** Cholesky: condition number $\kappa(C)$ vs. N ($H = 0.10$) with fitted power law.

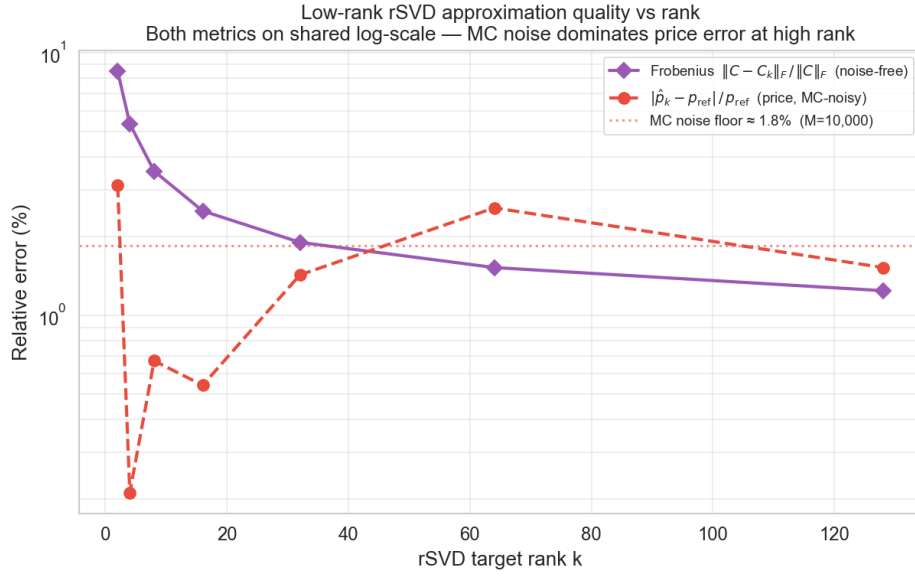


Figure 8: Low-rank rSVD approximation quality vs. rank k . Both the Frobenius error (noise-free, exact) and the price error relative to a 500,000-path reference are shown on shared log-scale axes. The dashed horizontal line marks the MC noise floor at $M = 10,000$ paths. Controlled: $N = 500$, $H = 0.10$. Independent: rank k . Dependent: Frobenius error and relative price error.

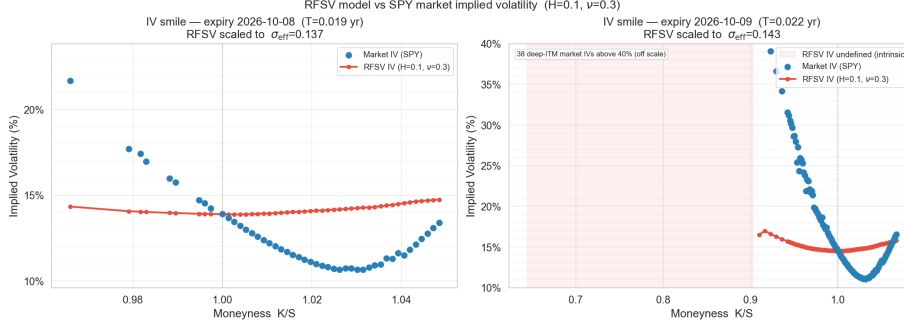


Figure 9: RFSV vs market implied volatility smile for SPY, for the two nearest expirations in a snapshot taken on October 1, 2026 (spot \$763.99). Scatter: market IV computed by Black–Scholes inversion of mid-quote (**brentq**, $r = 0$, filtered to strikes in $[0.60S, 1.40S]$ with valid bid/ask spread); the y -axis is capped at 40%, and the 38 deep-ITM market IVs above it are marked as off scale. Line: RFSV IV, same inversion applied to Monte Carlo European call prices, with common random numbers across strikes and the sample mean of S_T matched to the forward. The RFSV log-vol drift μ_0 is calibrated per expiration to match the ATM price (not IV): Brent’s method on the price residual $p_{\text{RFSV}} - p_{\text{mkt}}(\sigma_{\text{ATM}})$. The shaded region marks deep-ITM strikes where no simulated path finishes below K , so the Monte Carlo time value is zero and the IV is undefined. Controlled: $H = 0.10$, $\nu = 0.30$, $r = 0$, $M = 20,000$, $N = 63$ steps per year (at least 5 steps). Independent: strike K , expiration T . Dependent: implied volatility σ_{IV} .

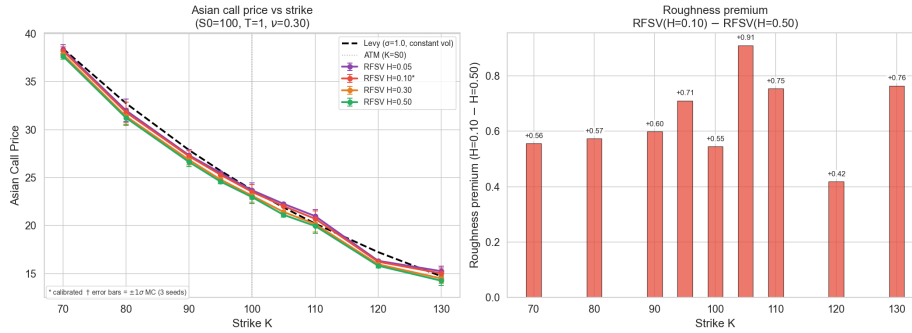


Figure 10: Asian option prices vs strike. **(a)** Price vs K : Lévy (1992) analytical approximation at $\sigma = 1.0$ (dashed, constant vol) and RFSV at four H values ($\nu = 0.30$); error bars are $\pm 1\sigma$ over 3 independent seeds, with the same seeds used for every H . **(b)** Roughness premium $\hat{p}(H=0.10) - \hat{p}(H=0.50)$ vs K , in option price units; peak near the money. Controlled: $S_0 = 100$, $T = 1$, $\nu = 0.30$, $r = 0$, $N = 252$, $M = 10,000$. Independent: K (both panels); H (left panel). Dependent: arithmetic Asian call price.

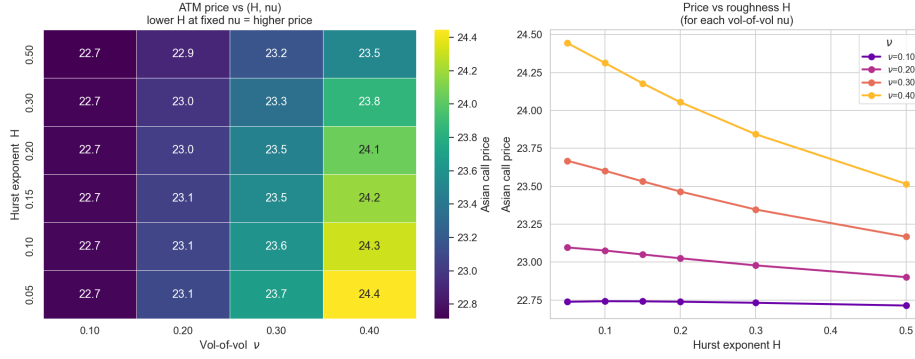


Figure 11: ATM Asian call price vs model parameters. **(a)** Heatmap: price vs (H, ν) at $K = S_0 = 100$ (viridis colormap; low H = rough = lower on the y -axis because the axis is inverted so H increases upward). **(b)** Price vs H for each ν ; curves decrease left-to-right and separate vertically with ν . All cells share the same random numbers, so differences between cells are precise even though each absolute price has an MC standard error of ≈ 0.6 . Controlled: $S_0 = 100$, $K = 100$, $T = 1$, $r = 0$, $N = 252$, $M = 10,000$. Independent: H , ν . Dependent: ATM Asian call price.

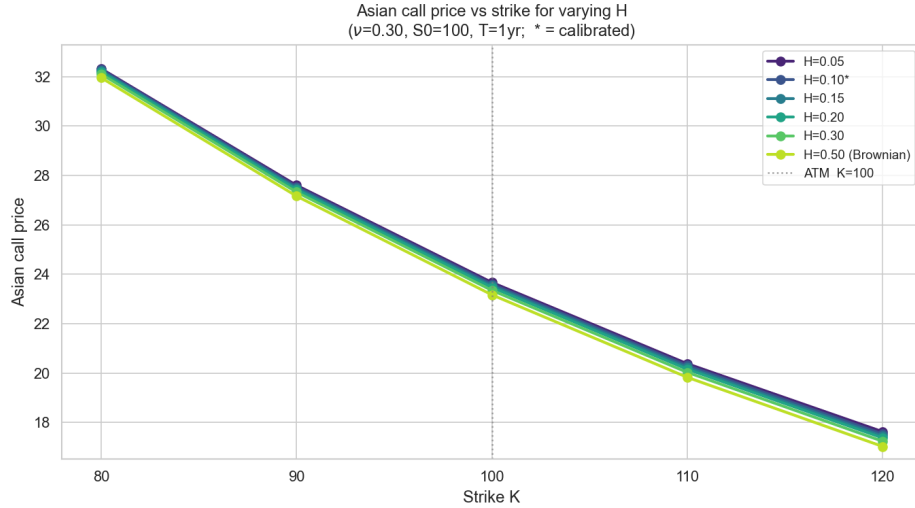


Figure 12: Asian call price vs strike K for varying H at $\nu = 0.30$, with common random numbers across all curves. Curves decrease monotonically with strike and are ordered by H at every strike; the spread between $H = 0.05$ and $H = 0.50$ widens from ≈ 0.3 at $K = 80$ to ≈ 0.6 at $K = 120$. Controlled: $\nu = 0.30$, $S_0 = 100$, $T = 1$, $r = 0$, $N = 252$, $M = 10,000$. Independent: K , H . Dependent: Asian call price.